



OPEN Plant leaf disease detection using vision transformers for precision agriculture

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Plant diseases cause major crop losses worldwide, making early detection essential for sustainable farming. Traditional methods need large training datasets, are expensive, and may overfit. In leaf image analysis, convolutional neural networks (CNNs) have revealed promise in leaf disease detection and classification. This research proposes PLA-ViT, or Precision Leaf Analysis with Vision Transformers, to improve agricultural monitoring. Vision Transformers (ViTs) outperform other neural networks because they employ self-attention to find global contextual information. The approach uses data augmentation, normalization, and bilateral filtering to increase generalization and image quality. Transfer learning using pre-trained ViTs reduces computing load and improves feature extraction. The model may be adjusted by hyperparameter tuning and adaptive learning rate scheduling for robust performance with minimal overfitting. In experiments, PLA-ViT outperforms other neural network-based models regarding detection accuracy, disease localization performance, inference time, and computational complexity. By attaching the system to IoT sensors, stakeholders may observe farms in real time and take timely measures like pesticide treatment or plant isolation. This novel method shows that transformer-based designs might help progress in precision agriculture.

Keywords Agriculture monitoring, Plant disease detection, Deep learning, Vision transformers, Classification

Early plant disease diagnosis is essential for producing healthy products¹. As humans use different kinds of plants to provide food, it is necessary to reduce the harmful effects of plant disease². Farmers can sometimes not accurately diagnose plant disease symptoms because of the tiny features³. Moreover, several farmers are not experts in diagnosing the disease, so artificial intelligence can help them provide better recognition⁴. Plant disease detection is done by looking at the leaf images taken using simple screening techniques⁵. On-site identification of diseases and pests of fruit trees is often made by agricultural and forestry professionals or farmers using their knowledge⁶. Nonetheless, it requires expertise and precision in observation, which is unavailable in some areas⁷. This approach is subjective, arduous, time-consuming, and ineffective⁸. Farmers with less knowledge could use medications carelessly and have errors during identification⁹. Environmental contamination brought on by quality and output will result in avoidable financial losses¹⁰.

Using deep learning to classify plant diseases has become an interesting field of study¹¹. Deep learning has reported remarkable performance in many computer vision applications to solve real-life problems that humans face daily, like medical image analysis and autonomous driving¹². In recent years, convolutional neural networks have been chosen for image processing as they have progressed in these fields¹³. Recently, the Vision Transformer (ViT) structure was introduced to improve classification applications¹⁴. The concept is based on how individuals categorize imageries. When a human observes a picture, they focus on a specific image area to detect the instance of interest¹⁵. The ViT design trails this method for picture classification. Using pre-designed architectures like ResNet, ViT, etc., is prevalent for computer vision tasks¹⁶. These models usually have many trainable parameters, requiring a large dataset to find the optimal values for their parameters¹⁷. Hence, these models are usually trained on a massive dataset like ImageNet, and the obtained weights are used for transfer learning¹⁸. A dataset with a domain identical to the first will gain high accuracy. This does not change that models would have difficulty getting satisfactory results on small datasets when the domain is different¹⁹. Additionally, the pre-designed designs negatively impact the prediction speed, which are often computationally intensive models.

The main contribution of the article is.

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- PLA-ViT methodology improves existing methodologies by employing multi-head self-attention mechanisms, data augmentation strategies, and adaptive learning rate schedules to improve the performance of the model.
- Furthermore, compared with other traditional CNN-based approaches, PLA-ViT promotes better feature learning due to the capability to acquire global contextual connections in leaves' images and then achieve higher accuracy in classification.
- Experimental evidence reveals that PLA-ViT achieved superior performance than other state-of-the-art deep models in disease localization accuracy, inference time, and computational cost.

The remainder of the article is as follows: Literature Study on Plant leaf disease identification in Section "[Literature survey](#)". Section "[Proposed method of precision leaf classification using CNN](#)" offers the proposed methodology. The models' numerical results are given in Section "[Results and discussion](#)", and lastly, Section "[Conclusion and future enhancements](#)" concludes our work with future work.

Literature survey

In this section the existing work relates to plant leaf disease detection is discussed with its merits and limitations. Previous and current research on using image processing to diagnose leaf diseases has received much attention and funding. There has been a rise in machine learning and image processing to identify agricultural diseases automatically.

Hybrid machine learning models

Santosh Kumar Sahu and Manish Pandey²⁰ suggested the hybrid random forest Multi-class Support Vector Machine (HRF-MCSVM) for plant leaf disease detection. Spatial Fuzzy C-Means were utilized to pre-process and segment the picture features before classification to improve computation accurateness. 54,303 images of healthy and diseased leaves comprise the Plant Village datasets. Lastly, the system's efficiency was measured using performance metrics like specificity, recall value, F1 score, accuracy, and sensitivity.

Monu Bhagat and Dilip Kumar²¹ recommended the Bag-of-words (BoW) and Speeded Up Robust Features (SURF) for leaf disease identification. Effectiveness, recall, F1-score, FNR, FPR, and MCC are metrics the suggested approach excels at achieving across all three datasets. On dataset1, dataset2, and dataset3, the proposed model achieved 97%, 97%, and 93% classification accuracy, respectively. The author has determined the percentage of feature reduction for the three species: pepper bell pepper, tomato, and potato, which is about 45.21%, 40.65%, and 34.73%, respectively.

Deep learning-based classification

Vivek Sharma et al.²² proposed the Deeper lightweight multi-class classification model (DLMC-Net) for plant leaf disease detection. The recommended model integrates a sequence of the collective block to extract deep features and the passage layer. The vanishing gradient issue is handled due to feature propagation and reuse advantages. The number of trainable variables is further decreased by utilizing separable convolution and point-wise blocks. On eight variables, error, accuracy, recall, precision, specificity, sensitivity, Matthews correlation coefficient and F1-score, the numerical outcomes of the suggested model are compared to those of seven existing models. With 92.34% accuracy on cucumber, 93.56% accuracy on citrus, 96.56% on tomato and 99.50% on grapes.

Towfek and Nima Khodadadi²³ discussed the Deep Convolutional Neural Network (DCNN) and Metaheuristic Optimization for Disease Detection in Plant Leaves. Several iterations of batch sizes, epochs, and dropout percentages were utilized to train the suggested model. Using validation information, the recommended model outperforms popular transfer learning approaches. According to extensive simulations, the suggested approach may achieve a staggering 83.12% accuracy in data categorization.

Yiping Chen et al.²⁴ presented cycle-consistent adversarial networks (CycleGAN) for leaf disease identification. Two discriminators are added based on CycleGAN: one to determine whether the picture is false or true and another to determine if the image of the leaf has disease. To maintain a balanced dataset for apple leaf diseases, healthy leaves are transformed into disease-carrying ones utilizing health-to-scab and health-to-rust models. In addition, the apple leaf disease is first trained using ResNet, a baseline CNN classifier. The findings demonstrate that ResNet achieves the greatest recognition accuracy on the test sets, attaining 97.78%. The created synthetic samples considerably enhanced the classification accuracy by 14.7%.

Vinay Gautam et al.²⁵ introduced the ensembled stack deep neural network (ESDNN) for mango leaf disease classification and detection. This effort aims to create an effective and effective AI-based system for early disease recognition and classification of leaves. The most obvious approach to avoid losing time and money is to identify crop diseases earlier. The primary use of the plant leaf picture is the diagnosis of plant diseases. Automated disease recognition in mango leaves is addressed in this work using an ESDNN model. The recommended model achieves an accuracy rate of 98.57%, greater than existing models.

Object-detection and real-time systems

JanibulAlam Soeb et al.²⁶ deliberated the YOLOv7 for Tea leaf disease detection and identification. The problem of inadequate sample sizes is addressed in this work by using data augmentation methodologies. Notable statistical indicators such as prediction accuracy (97.3%), recall (96.4%), precision (96.7%), mAP value (98.2%), and F1-score (0.965) verify the identification and detection findings for the YOLOv7 technique. Comparing YOLOv7 to other target recognition and identification networks, such as Deep CNN, CNN, DNN, AXRetina Net, improved YOLOv5, DCNN, and Multiobjective picture segmentation, the experimental findings show that YOLOv7 is the best for tea leaf disease in natural scene photos.

SAKIB HOSSAIN SHOYON et al.²⁷ examined the Multi-Model Ensemble Method (PlantDet) for Plant Disease Detection. For datasets with limited background images, PlantDet addresses underfitting issues and concurrently leverages nourished performances. Compared to all existing models, PlantDet is more robust thanks to its effective data augmentation and pre-processing, Dropout mechanism, Pooling layer, PReLU activation function, L2 regularizers, Dense layers and Batch Normalization layers. When tested on the Rice Leaf dataset, PlantDet achieved better outcomes than the previous existing model concerning Accuracy (98.53%), Recall (98.35%), Precision (98.50%), Specificity (99.71%) and F1 (98.42%).

Xibei Huang et al.²⁸ investigated the Fully Convolutional-Switchable Normalization Dual Path Networks (FC-SNDPN) for the Tomato Leaf Disease Detection System. An FCN based on the VGG-16 models was utilized to segment the crop picture, mitigating the impact of the intricate background on the recognition of insect crop and pest diseases. Then, enhanced Dual-Path network models are suggested to enhance the feature extraction ability.. The suggested technique, which uses FCN for foreground segmentation and SNDPN for detection, achieves an impressive detection accuracy of 97.59% on the augmented data set.

Advanced techniques and ensemble methods

Senthil Pandi Sankarshwaran et al.²⁹ suggested that the crossover boosted artificial hummingbird algorithm-based AX-RetinaNet (CAHA-AXRNet) for detecting rice plant disease. The key focus of this research article is the efficacy of disease detection and categorization in rice plants. The CAHA optimization models were used to optimize the AX-RetinaNet model's hyperparameters. To confirm the success of disease detection, the most critical performance indicators are recall, specificity, accuracy, F1-score, and precision. With an accuracy rate of 98.1%, the suggested CAHA-AXRNet technique outperforms other approaches currently used to identify diseases in rice plants.

Vivek Sharma et al.³⁰ proposed the Crop Leaf Gan (CLGan) and deep CNN for maize leaf disease identification. The encoder-decoder network has been encased in the CLGan's generator and discriminator to maintain complex particulars during synthetic image production with substantial lesion differentiation. This helps to avoid issues like training instability, non-convergence failure, and the vanishing gradient problem. The suggested enhanced loss function maintains efficient weight optimization and stabilizes learning with a dynamic correction factor. While using the fewest parameters, CLGanNet attained the greater experimental results, with a training accuracy of 99.97% and a testing accuracy of 99.04%. Table 1 displays the summary of the existing methods.

The input image classification results from deep learning algorithms are encouraging, yet differ for a variety of reasons, including data imbalance and the presence or lack of attributes across dataset classes³¹. Tomato plant disease categorization using Multilevel Feature Fusion Network (MFFN) is suggested in this study. For the purpose of classifying photos of tomato leaves from plants, it makes use of ResNet50, MFFN, and the Adaptive Attention Mechanism, which integrates channel, spatial, and pixel attention. An outstanding 99.88% training accuracy, 99.88% validation accuracy, and 99.83% external testing accuracy were attained by the proposed deep learning-based technique when evaluated on a tomato plant leaves dataset.

When addressing data imbalances and similar symptoms, it is essential to employ a new method for accurate plant disease identification that is based on deep learning. Incorporating adaptive attention techniques, the proposed Multilevel Feature Fusion Network (MFFN) takes account of a wide variety of network buildings, making it more resilient³². The network achieves outstanding results on photos with complicated backgrounds using this benchmark dataset that was built with a complex background and cardamom plant disease classification using U-Net for background removal and EfficientNetV2 for classification. With a 99% success rate

Reference	Environment	Algorithm	Accuracy	Scalability	Limitations
Sahu and Pandey ²⁰	Machine Learning	Optimal Hybrid Multi-Class SVM with Spatial Fuzzy C-Means	96.5%	High	High computational cost due to hybrid approach; requires large datasets for training
Bhagat and Kumar ²¹	Machine Learning	BoWs and SURF for Feature Selection	94.3%	Low	Feature selection is sensitive to noise and lacks real-time processing capabilities
Sharma et al. ²²	Deep Learning	DLMC-Net (Deeper Lightweight Multi-Class Classification Model)	97.2%	Moderate	Limited generalization for unseen plant diseases may require further model optimization
Towfek and Khodadadi ²³	IoT & Edge Computing	Deep CNN with Metaheuristic Optimization	95.8%	High	Computationally expensive, metaheuristic optimization may not always converge optimally
Soeb et al. ²⁶	Computer Vision	YOLOv7 (YOLO-T) for Tea Leaf Disease Detection	98.1%	Moderate	Performance depends on dataset quality; struggles with small and complex disease regions
Chen et al. ²⁴	Generative AI	Improved CycleGAN and CNN for Apple Leaf Disease Identification	96.0%	Low	Requires high-quality synthetic data; potential overfitting on smaller datasets
Gautam et al. ²⁵	Machine Learning	ESDNN (Ensembled Stack Deep Neural Network) for Mango Leaf Disease Detection	97.5%	High	High memory and computational demands; ensembling increases training complexity
Huang et al. ²⁸	IoT & Smart Agriculture	FC-SNDPN for Tomato Leaf Disease Detection	95.7%	High	Lacks adaptability to varying environmental conditions; requires extensive pre-processing
Shoyon et al. ²⁷	Deep Learning & IoT	PlantDet (Multi-Model Ensemble Method)	98.3%	Moderate	Ensemble methods increase inference time and may require a large dataset for optimal performance
Sankarshwaran et al. ²⁹	Edge AI & IoT	AX-RetinaNet with Crossover-Boosted Artificial Hummingbird Algorithm	96.8%	Low	High computational cost; limited interpretability of the optimization process
Sharma et al. ³⁰	Blockchain & AI	CLGanNet (CLGan and Deep CNN) for Maize Leaf Disease Identification	97.0%	High	Training is computationally intensive and requires significant hyperparameter tuning for optimal performance

Table 1. Summary of the existing models.

on the tomato plant dataset and a 98.28% success rate on the cardamom leaf dataset, this study demonstrated promising outcomes.

Various methods for plant disease detection or classification based on Deep Learning and Machine Learning are presented in this systematic review³³. The study takes into account 160 separate research papers, and the models included range from single network models to hybrid models and even real-time detection methods. About 57 research looked at different plants, whereas 103 materials focused on only one. They include 50 distinct datasets on plant leaf diseases, including both publicly available and privately held databases.

Most previous research on disease detection in plant leaves has relied on ML/DL hybrid techniques, such as CNNs, SVMs, YOLO, and ensemble methods. Research in this area uses metaheuristic optimizations, feature extraction methods such as BoWs and SURF, and other similar approaches to improve classification precision. Problems with generalization to new diseases, susceptibility to noise, enormous dataset reliance, and high computing costs remain. Certain models aren't as successful regarding large-scale implementation in IoT-based smart agriculture since they can't adapt to real-world situations. This study presents Precision Leaf Analysis with Vision Transformers (PLA-ViT), a new framework that uses Vision Transformers (ViT) to improve feature extraction and disease classification to overcome these difficulties. In contrast to CNN-based models, PLA-ViT guarantees more accuracy and flexibility by accurately capturing complex spatial linkages and long-range dependencies in pictures of leaves. Additionally, the model is more interpretable and less noise-resistant because of its transformer-based design. One major improvement over previous methods is the capability to monitor diseases in real-time with less computing overhead via the integration of PLA-ViT into an Internet of Things (IoT)-based smart agricultural ecosystem.

Proposed method of precision leaf classification using CNN

The methods and resources that were accessible to us during our research are detailed in this section. This article presents the primary dataset that underpins our analysis by outlining its make-up and preparation procedures. As a further step forward in computer vision, this research thoroughly examines the novel ViT architecture. Promising new directions for feature extraction and pattern detection, the ViT stands out by efficiently capturing long-range relationships in picture data via self-attention processes. As a prerequisite to grasping the paper's analyses and conclusions, this methodological introduction provides background information.

By leveraging on Vision Transformers' (ViTs) ability to understand both global and local spatial interactions, the suggested Precision Leaf Analysis with Vision Transformers (PLA-ViT) framework overcomes significant shortcomings in current plant disease detection approaches. By utilizing multi-head self-attention processes, PLA-ViT improves disease localization and classification, in contrast to conventional CNN-based methods that depend on local receptive fields and could overlook subtle spatial relationships. By lowering computational cost and improving the model's accuracy in recognizing fine-grained illness patterns, this improvement makes it appropriate for real-time deployment on devices with limited resources. Moreover, PLA-generalization ViT's performance on various and unbalanced leaf datasets is improved by the integration of data augmentation, adaptive learning rate scheduling, and sophisticated pre-processing approaches. In terms of computational complexity, inference time, and detection accuracy, experimental comparisons show that PLA-ViT surpasses state-of-the-art models like DLMC-Net, CycleGAN, and ESDNN. This helps advance precision agriculture by providing faster, more scalable, and highly accurate disease detection.

Problem statement

The adverse effects of plant leaf diseases on agricultural output and quality are a major concern on a worldwide scale. Global food security is at risk because these diseases lower crop yield efficiency. Manual observation and expert diagnosis, the two most common traditional disease detection techniques, are unsuitable for large-scale agricultural operations because of their high costs, high labour requirements, and lengthy detection times. Moreover, traditional methods often miss early-stage disease signs, which increases crop losses and decreases the efficacy of disease control. Automated detection approaches sometimes employ models that are too big for low-resource devices. Small datasets make it difficult to create reliable models to diagnose plant diseases in various settings. Currently, the present techniques cannot reliably and quickly identify illnesses. Delaying action permits agricultural damage to persist despite the necessity for early detection. The lack of research on machine learning-based plant disease diagnostic solutions contributes to the lack of effective and scalable disease detection systems. Separating health issues from dietary shortages is another challenge that sometimes leads to erroneous diagnoses. Rural farmers lack the technology to handle disease outbreaks, which hinders the broad use of automated disease detection systems. To quickly solve these difficulties, plant disease detection systems that are automated, accurate, scalable, and easy to install on low-resource devices are needed. These tools help farmers detect diseases early and gather useful data. This enhances sickness management and enables them to intervene when needed. Even if deep learning methods have improved, overfitting, constrained datasets, and the necessity for efficient models make it challenging to produce an effective and accessible solution. The system for early detection of plant leaf diseases must be improved by integrating cutting-edge models and overcoming the limits of present approaches. Contemporary machine learning methods like ViT and advanced CNN may improve plant disease diagnosis and categorization in complex environments. Foliar diseases threaten agricultural productivity. This study will discuss these challenges and try to find a solution to improve plant disease identification in the following paragraphs. This study will boost agricultural productivity and greener plant-growing methods.

Although Convolutional Neural Networks (CNNs) and other conventional deep learning models have shown impressive performance on image-based tasks, the Visual Transformer (ViT) has a number of benefits that make it the superior choice. A significant benefit is that it performs effectively on bigger datasets since it can more effectively capture long-range relationships. The ViT models global context without hierarchical feature

extraction, in contrast to CNNs that depend on local receptive fields. It does this by using self-attention processes to examine entire patches sequentially.

Furthermore, when fine-tuned for particular applications and pre-trained on large data sets, the Visual Transformer demonstrates improved scalability and generalizability. With its adaptability, it can handle different input distributions and prevent overfitting in situations with various image inputs. When it comes to computational efficiency on hardware tuned for transformer-based architectures, interpretability of feature maps, and resilience to adversarial attacks, ViTs frequently exceed CNNs, even when the accuracy is equivalent. Because of their improved handling of complicated spatial connections in images, their capacity to scale with data, and their architectural simplicity, ViTs are becoming more and more popular, even if it's just as accurate.

Figure 1 shows the proposed PLA-ViT Model. Input data collection starts with capturing high-resolution photographs of plant leaves using various means, such as drones, smartphone cameras, and leaf scanners based on the IoT. This ensures that the datasets are varied and of good quality. To make the model more stable, these pictures are improved at the pre-processing step by adjusting the contrast, reducing the noise, and adding data (rotating, flipping, and varying the brightness). The input dimensions for the Vision Transformer (ViT) model are also standardized via scaling and normalization. Utilizing ViT, the feature extraction step converts input pictures into a series of fixed-size patches, each flattened into a vector representation, by patch embedding. Because conventional CNNs only use local receptive fields, ViT can learn complex patterns from several disease locations thanks to its multi-head self-attention mechanism, which captures local and global spatial dependencies. The model also incorporates positional embeddings into the patch embeddings to ensure that the model preserves spatial information about the leaf structures, which is necessary since transformers do not have natural spatial awareness. To differentiate between various plant diseases like rust, blight, and mildew and award confidence ratings, the feature vectors extracted are processed via a refined ViT model in the disease classification module. In addition to classification, attention heatmaps, bounding boxes, and segmentation methods are used for disease localization and severity analysis to highlight infected locations and determine mild, moderate, or severe degrees. The decision support system offers practical insights through the recommendation of pesticides or fungicides, the generation of crop health data, and the delivery of notifications to farmers via the IoT. Lastly, systems for intervention and monitoring allow for tracking diseases over extended periods. These systems include autonomous drone spraying and storing previous data in the cloud.

Table 2 summarizes the vision transformer model. The PatchEncoder layer uses a sliding window technique to extract patches from 16×16 pixel pictures. A linear projection to each patch and positional embeddings to the data adds spatial information. This process produces many patch embeddings. After that, the model has many TransformerEncoder layers with position-wise feedforward networks and multi-head self-attention processes. Layer normalization and residual connections sustain transformer layers. Layers are needed to capture local and global dependencies in the image. The scalable and flexible model includes adjustable parameters like patches, hidden size, and transformer heads. At a learning rate 0.0001, the Adam optimizer adjusts model parameters during training to guarantee efficient convergence. Removing encoded information and processing them with deep layers improves the model's ability to learn complicated patterns. Finally, the model's output layer uses SoftMax activation to properly identify input photos by predicting output classes. The ViT model's impressive classification of image performance shows its versatility in visual recognition domains. This was achieved via intensive labelled image data training and parameter optimization.

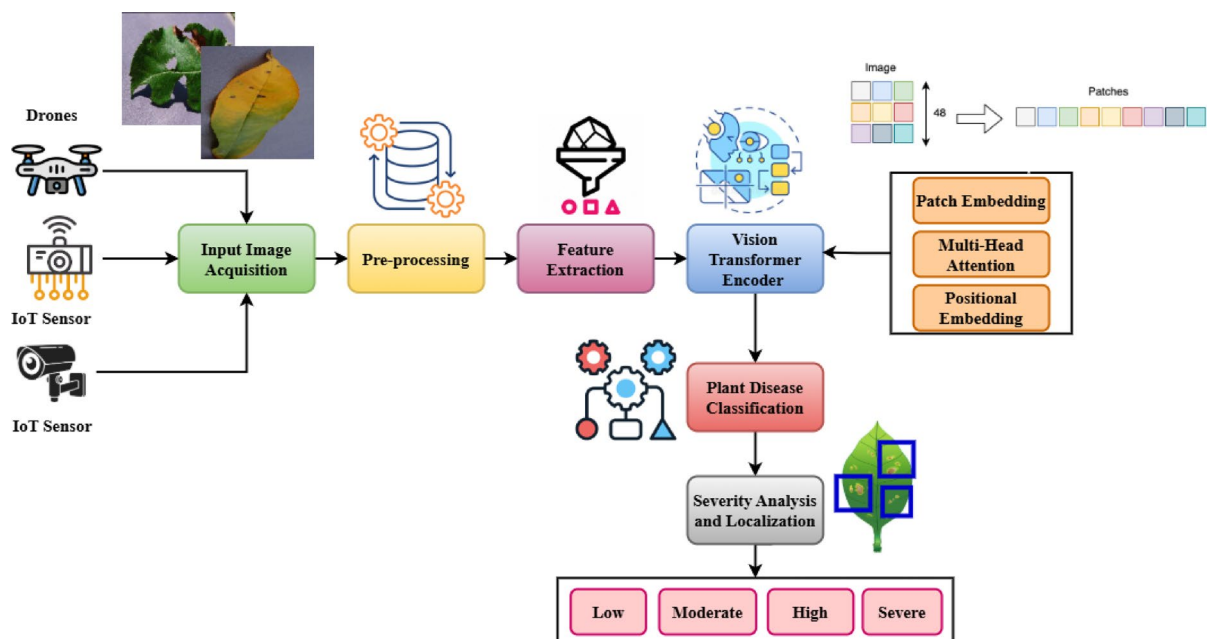
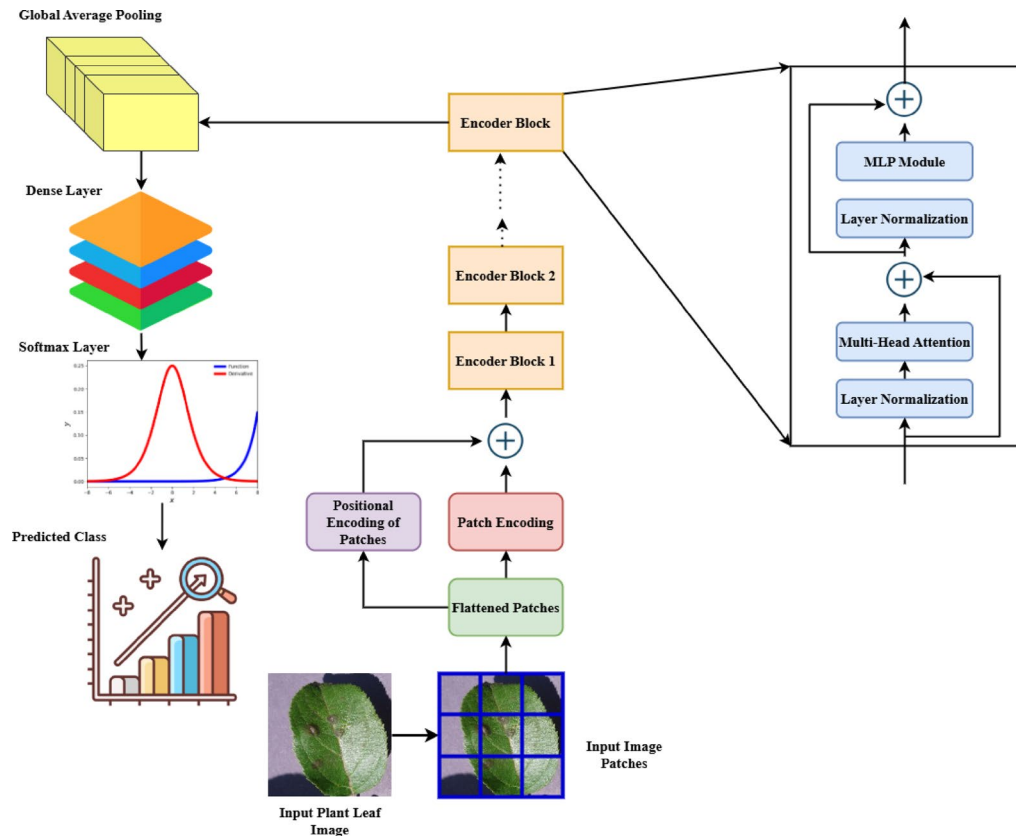


Fig. 1. Precision leaf analysis with vision transformers (PLA-ViT).

Layer (type)	Output shape	Parameters
Patch embedding (Conv2D)	(Batch, 196, 768)	590,592
Position embedding (Learnable)	(Batch, 197, 768)	151,296
Transformer encoder (12 layers)	(Batch, 197, 768)	87,032,320
Multi-head self-attention (MSA)	(Batch, 197, 768)	2,362,368 per layer
Feedforward network (MLP)	(Batch, 197, 768)	9,437,184 per layer
Layer normalization	(Batch, 197, 768)	1,536 per layer
CLS token (Learnable)	(Batch, 1, 768)	768
MLP head (Classification Layer)	(Batch, 1000)	769,000

Table 2. Summary of vision transformer model.**Fig. 2.** ViT network architectural design.

Patch generation and positioning encoding

Figure 2 shows the ViT network architectural design. It includes multiple encoder blocks linked by direct connections only. Every ViT model encoder block comprises Multi-Layer Perceptrons (MLP), Layer Normalization and Multi-Head Attention modules. Vision Transformers (ViT) takes the 2D image $I \in \mathbb{R}^{h \times w \times C}$ and partition it into non-overlapping patches of size Q . The picture endures a transformation progression, resulting in patch sequences signified as $J_q \in \mathbb{R}^{M \times (Q^2 \cdot C)}$, where Q indicates patch sizes, and M denotes the number of patches computed as $M = (h \times w)/Q^2$. The procedure of embedding includes taking input patches, flattening them, and projecting them into D -dimension spaces utilizing linear projections $E = [J_q^1, J_q^2, \dots, J_q^M]$. The patch embedding is combined with the class embeddings $y_{class} \in \mathbb{R}^{1 \times D}$. Position embedding is merged into the joined embedding to reserve the positional particulars.

$$E_{POS} = \begin{cases} \sin\left(\frac{pos}{1000 \cdot \frac{2j}{d}}\right), & \text{if } j \text{ is even} \\ \cos\left(\frac{pos}{1000 \cdot \frac{2j}{d}}\right), & \text{if } j \text{ is odd} \end{cases} \quad (1)$$

The embedding process is signified by the Eq. (2).

$$Y_j = [y_{class}; J_q^1, J_q^2, \dots, J_q^M] + E_{POS} \quad (2)$$

As shown in Eq. (2), where $y_j \in \mathbb{R}^{(M+1) \times D}$ signifies the last embedding for j th patch and $E_{POS} \in \mathbb{R}^{(M+1) \times D}$, D denotes the transformer's latent vector sizes. Algorithm 1 shows the precision leaf classification using a convolutional neural network.

Step 1: Initialize Parameters

1. Call initialize_weights()

Step 2: Training Process (Per Epoch)

For each epoch:

- For each image in the training dataset:
 1. Convolutional Layer:
 - Compute convolution:
conv_output = convolution(image, kernel)
 2. ReLU Activation:
 - Apply activation function:
relu_output = relu(conv_output)
 3. Pooling Layer:
 - Apply max pooling:
pooled_output = max_pooling(relu_output)
 4. Fully Connected Layer:
 - Process data through the fully connected layer:
fc_output = fully_connected(pooled_output)
 5. Softmax Layer:
 - Apply softmax for classification:
predictions = softmax(fc_output)
 6. Compute Loss:
 - Calculate loss based on predictions and true labels:
loss = compute_loss(predictions, true_labels)
 7. Backward Propagation & Weight Update:
 - Compute gradients:
gradients = backward_propagation(loss)
 - Update model weights:
update_weights(gradients)

Step 3: Function Definitions

- The following functions are implemented:
convolution(), relu(), max_pooling(), fully_connected(), softmax(), compute_loss(),
backward_propagation(), update_weights()

Algorithm 1. Precision leaf classification using convolutional neural networks.

The suggested approach does early global feature inference in stage 1 using ViT. Splitting the input picture into predefined-size patches transforms it into a linear input sequence. We apply linear and position

embedding to every patch and then feed the standard Transformer encoder the resultant linear input sequence. Supplemental classification tokens are appended at the beginning of the series before training to achieve picture classification. MLP and Multi-head Self-Attention (MHSA) are components of the Transformer encoder. All modules use LayerNorm (LN) to normalize and include residual connections. The computations carried out by each component are represented by Eqs. (3) to (5).

$$Z_{0l'} = MHSA(LN(Z_{l-1})) + Z_{l-1} \quad (3)$$

$$Z_l = MLP(LN(Z_{0l'})) + Z_{0l'} \quad (4)$$

$$x = LN(Z_l^0) \quad (5)$$

Each CNN layer's characteristics are location, 2-D neighbourhood structures, and shift-invariance. Unless otherwise mentioned, it performs operations on a range of real values. Though the MLP layer is the only one in ViT to display locality and shift invariance, the self-attention layer seizes global characteristics. Stage 1 employs ViT for global feature inferences. In contrast to CNN, ViT quickly extracts early-stage characteristics advantageous to the network and emphasizes global features. Over time, the number of features rises as the network deepens. By using the ViT module in Stage 2, global feature modelling and parameter reduction in the network are implemented. It may express these processes using Eq. (5).

$$T = HW_{softmax}(YW_A)^T Y \quad (6)$$

As inferred from Eq. (6), where $YW_A \in R^{C \times L}$ forms semantic group from Y , $SoftMax(\cdot)$ denotes the softmax activation functions, and Y signifies feature maps. For the input feature maps Y , ViT utilizes point convolution to maps every pixel $Y_p \in R_c$ in the feature maps into L group, then utilizes spatial pooling to determine the token. ViT adopts a concatenation approach to discourse the high-level semantic data sparsity and evade computational leftovers.

Figure 3 shows the overall flowchart of the suggested model. the sequential processing of the proposed PLA-ViT model, from data acquisition to final classification. A crucial part of the suggested microgrid design, the hybrid PV/Wind/Battery system is illustrated in Fig. 3 in a comprehensive schematic layout. The interaction between the Battery Energy Storage System (BESS) and renewable energy sources (RES) like wind turbines and solar panels is emphasized, and how this interaction ensures a constant supply of energy and makes the system more resilient to variations in RES output. The control method that optimizes energy dispatch while accounting for uncertainties in solar and wind power generation relies on this schematic, which is vital for understanding the underlying functions. The operating dynamics of the microgrid and the significance of combining distributed generation (DG) units with energy storage to enable cost-efficient, demand-side management (DSM)-enhanced scheduling could be enhanced with the assistance of Fig. 3, which visualizes the connections and energy flows.

The first steps in training the ViT model include gathering and pre-processing picture data to ensure consistency in the dataset. After that, the hyperparameters are defined, which include learning rates, batch sizes, and optimization methods to improve the model's performance. Next, implement Shifted Patch Tokenization. This module preserves spatial information by segmenting pictures into overlapping patches. Visualizing the patches lets us assess their structure and input data quality. Patch encoding modules encode patches into transformer-friendly token embeddings after validation. The Locality Self-Attention module will reveal image data's global and local interdependence. Next, the MLP module improves feature extraction and representation learning. After integrating these components, the Vision Transformer model is built by organizing the many modules into a framework. The model is put through its paces through compilation, training, and assessment to achieve optimum performance by repeated validation and learning. The last step is to analyze the trained model's output, which includes predictions, to see how well it recognizes patterns and characteristics in the dataset. Finally, the outputs are evaluated to ensure the ViT model meets the necessary performance standards. The convolutional block attention module block comprises channel attention modules and spatial attention modules, and it utilizes 3×3 convolutional kernels to pre-process feature maps earlier insertions. This module passes the input feature maps via parallel max pooling layers and average pooling layers, and then this study changes feature maps from $[C, H, W]$ to $[C, 1, 1]$ dimension. The common MLP modules comprise a 1×1 convolutional kernel that reduces the channel count to R epochs its initial value and then restores it to its initial value. The original feature map is multiplied by the weight of every channel, which is created by splicing the feature map acquired by the max pooling and average pooling layers. Equation 7 explains how the channel attention module assigns weights. ρ denotes for utilizing Sigmoid as activation functions, $\omega_0 \in \mathbb{R}^{C/r \times C}$, and $\omega_1 \in \mathbb{R}^{C/r \times C}$. ω_0 and ω_1 are shared weights for the 2 inputs of the average and max pooling layers, as in Eq. (7).

$$\begin{aligned} N_c(f) &= \rho(MLP(Avgpool(f))) + MLP(Maxpool(f))) \\ &= \rho(\omega_1(\omega_0(f_{avg}^c)) + \omega_1(\omega_0(f_{max}^c))) \end{aligned} \quad (7)$$

The average and max pooling layers determine the channel attention module output. This study acquires 2 feature map with a dimension of $[1, H, W]$ and then splices them. Over and done with 7×7 convolutions, this study obtains feature maps of one channel and multiples it by actual feature maps. To evaluate the importance of every ViT's output, this study introduces learnable attention weight vector ω_j . The attention scores b_j is computed by the dot products of the ViT's depiction g_j and the attention to weights ω_j as in Eq. (8):

$$b_j = g_j^T \omega_j \quad (8)$$

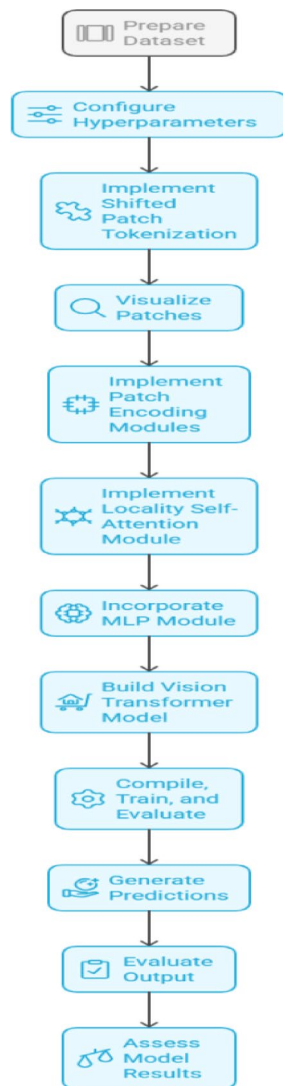


Fig. 3. Sequential flow of the suggested model.

The calculated attention score is then standardized utilizing softmax functions to generate normalized attention weight β_j , signifying the relative significance of every ViT. The standardization is executed in Eq. (9):

$$b_j = \frac{\exp(\beta_j)}{\sum_{i=1}^N \exp(\beta_i)} \quad (9)$$

As shown in Eq. (8), where $N = 5$ is the number of ViTs utilized. The standardized attention weight β_j reflect each ViT's input to the last forecast. This research uses these normalized attention weights to compile the several ViT representations into a single final representation. Equation (10) shows the weighted summation that is used to produce the aggregation:

$$G = \sum_{j=1}^N \beta_j g_j \quad (10)$$

The suggested PLA-ViT confusion matrix for all the datasets included is shown in Fig. 4. The dark blue diagonal shows true positives (TP), and correct classifications are shown by the saturation. While the existing models and the suggested PLA-ViT make occasional mistakes, the former shows better overall classification accuracy. The complex qualities of the input picture are better reflected in this aggregated representation, which incorporates the varied information received from numerous ViTs. This research uses a linear transformation layer, defined mathematically in Eq. (11), to improve this composite representation:

$$Gt = \text{Linear}(G) \quad (11)$$

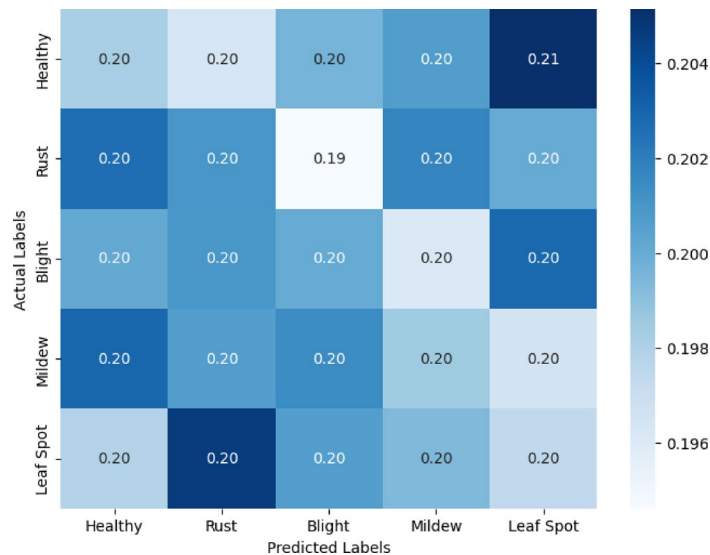


Fig. 4. Confusion Matrix of the proposed PLA-ViT for all the included datasets.

The refined illustration G' is then passed to the classification head to produce the last logits. The model can deliver a likelihood distribution across preset classes since the classification head is usually fully connected to layer with softmax activation functions for multi-class classifications. By including varied representations from many ViTs, this method seeks to increase the model's generalizability and overall effectiveness. By integrating data from several sources, the model is better equipped to deal with intricate image patterns and variances, leading to more precise classification. Algorithm 2 shows the Precision Leaf Analysis with Vision Transformers.

Step 1: Initialize Parameters

1. Call `initialize_transformer_weights()`

Step 2: Image Pre-processing**1. Split Image into Patches:**

- `patches = split_image_into_patches(image)`

2. Linear Projection of Flattened Patch:

- `projected_patches = [linear_projection(patch) for patch in patches]`

3. Add Position Embeddings:

- `position_embeddings =`
`create_position_embeddings(len(projected_patches))`
- `encoded_patches = projected_patches + position_embeddings`

Step 3: Transformer Encoder Layers**• For each layer in the transformer:****1. Multi-Head Self-Attention:**

- `attention_output = MHSA(encoded_patches)`

2. Add & Normalize:

- `encoded_patches = layer_norm(attention_output + encoded_patches)`

3. Feedforward Network (MLP):

- `mlp_output = mlp(layer_norm(encoded_patches))`

4. Add & Normalize:

- `encoded_patches = layer_norm(mlp_output + encoded_patches)`

Step 4: Classification Head**1. Process Output Through Classification Layer:**

- `class_output = classification_head(encoded_patches)`

Step 5: Compute Loss and Backpropagation**1. Compute Loss:**

- `loss = compute_loss(class_output, true_labels)`

2. Perform Backward Propagation and Update Weights:

- `gradients = backward_propagation(loss)`
- `update_transformer_weights(gradients)`

Algorithm 2. Precision leaf analysis with vision transformers.

PLA-ViT improves feature extraction and disease classification by prioritizing local spatial hierarchies over CNN-based models. This utilizes long-range dependencies in plant leaf pictures instead of CNN-based models. A pipeline that pre-processes and supplements data is advised to improve model generalizability. Vision Transformer's multi-scale feature extraction technology helps grasp complicated sickness patterns. After that,

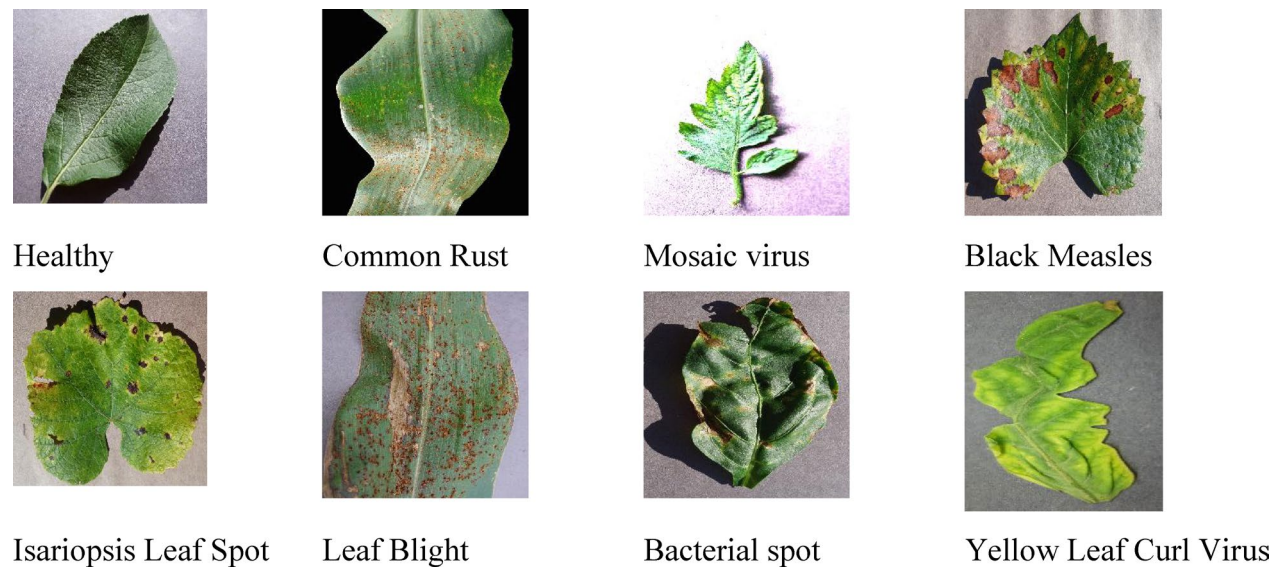


Fig. 5. Examples of the plant leaf disease dataset.

Parameter	Value
Input size	256 × 256
Patch size	16 × 16
Weight decay	0.0001
Batch size	256
Learning rates	0.001
Optimizers	Adam
Dropout ratio	0.2
Learning rate reduction	0.5 to 15 epoch
Validation loss thresholds	0.001

Table 3. Hyperparameters and their value for the ViT model’s operation.

the classification and identification module can accurately identify the ailment and its severity. This study used model pruning and lightweight attention approaches to increase computational efficiency. These improvements cut inference time without affecting accuracy. The PLA-ViT model enhances accuracy, sickness localization, inference time, and calculation complexity compared to previous models.

Results and discussion

This study describes the Precision Leaf Analysis with Vision Transformers (PLA-ViT) system for autonomous leaf disease prediction and precision agriculture monitoring. The PLA-ViT model was compared to HRF-MCSVM, DLMC-Net, CycleGAN, and ESDNN in accuracy, illness localization, inference time, and computational complexity.

Dataset description

CNN has been used in numerous works to categorize diseases because it can learn relevant features. Drawbacks caused by various model architectures, size of the dataset, imbalance dataset, leaf image categories like low resolution, environment condition, and real-time images. There are certain diseases that have similar features in such instances; the model miss-classify input images. The performance of the classification model is affected by numerous factors, including selection of the plant leaf images essential attributes; it significantly impacts imbalances and small datasets with fewer spatial pathogen characteristics³⁴. Figure 5 shows examples of the plant leaf disease dataset.

Experimental environment

A cloud platform based on deep learning runs all the trials. The system specs include 45 GB RAM, an NVIDIA GeForce RTX 3090 GPU, a 14-Core VV Intel(R) Xeon(R) Gold 6330 CPU running at 2.00 GHz, and more. Ubuntu 18.04 is the OS, while the software stack consists of Python 3.8 and PyTorch 1.9.0. Table 3 shows the Hyperparameters and their value for the ViT model’s employment. Conventional deep learning architectures, including CNNs, are based on local feature extraction using convolutional operations, which restrict their

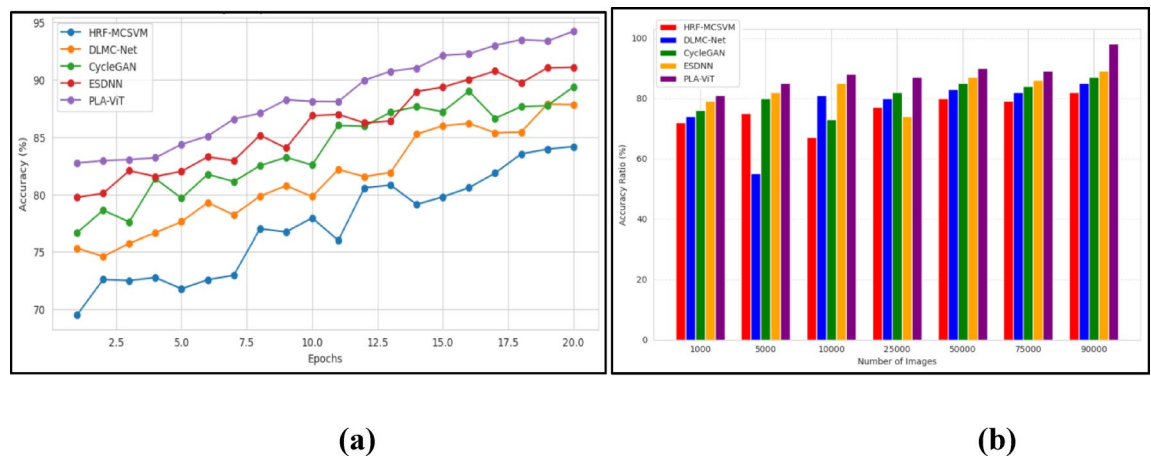


Fig. 6. Accuracy Ratio based on (a) Epochs and (b) Number of Images.

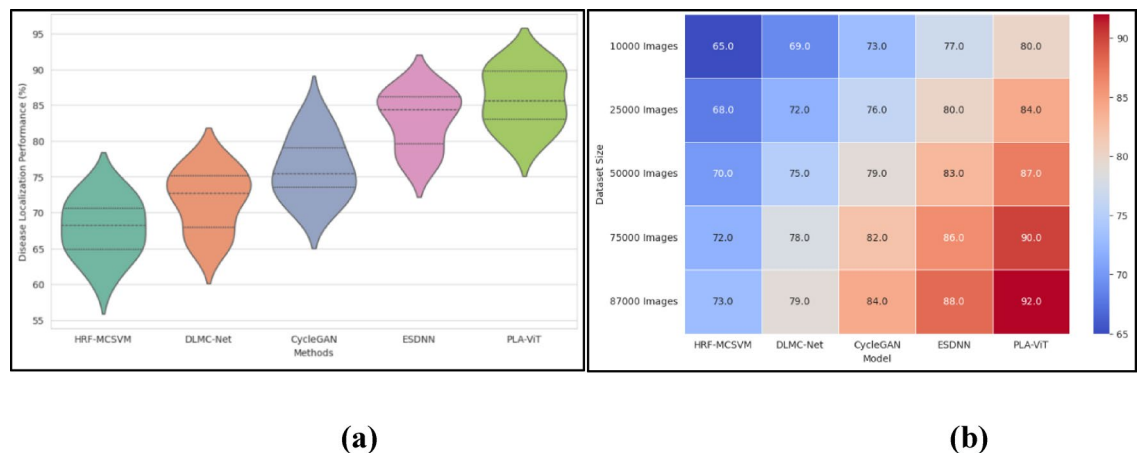


Fig. 7. Disease localization performance analysis.

capacity to learn long-range dependencies. Vision Transformers (ViTs) use self-attention mechanisms to extract global contextual information, improving disease classification and localization. Although CNNs provide high accuracy, ViTs provide better interpretability, flexibility, and scalability, which make them useful for precision agriculture applications.

Accuracy ratio

A Transformer and densely linked networks will be the fundamental components of a hybrid model that this research anticipates can achieve tiny parameters while providing excellent classification accuracy. Although convolutional neural network (CNN) models excel at extracting local information, their limited perceptual fields make them struggle when capturing localized data representations. Yet, the Transformer's built-in self-attention mechanism excels at collecting contextual linkages and long-range dependencies but struggles with localized, fine-grained information. Maximizing global and regional feature information retention may be achieved by combining a convolutional network with a Transformer. This combination offers many benefits. Figure 6 and Eq. (12) shows the accuracy ratio.

$$Accuracy = \frac{TruePositive + TrueNegative}{TruePositive + TrueNegative + FalsePositive + FalseNegative} \quad (12)$$

Disease localization performance

Deep learning methods have been very successful; however, they tend to tackle disease localization and classification independently. Plant disease detection may be more thorough and practical by combining localization and classification activities into a multi-tasking framework. The suggested multi-task learning approach seeks to optimize the accuracy and efficacy of disease recognition in plant leaf pictures by concurrently improving the disease classification and localization tasks. The goal is to use shared knowledge across the tasks to perform better. The model can accurately localize diseases by capturing crucial patterns and correlations using

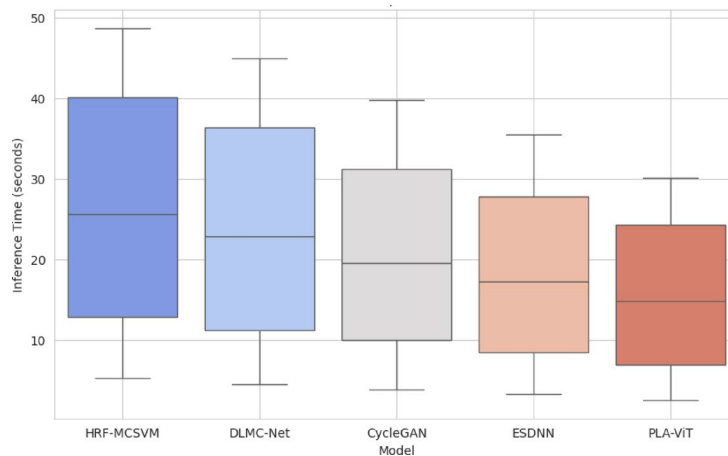


Fig. 8. Inference time.

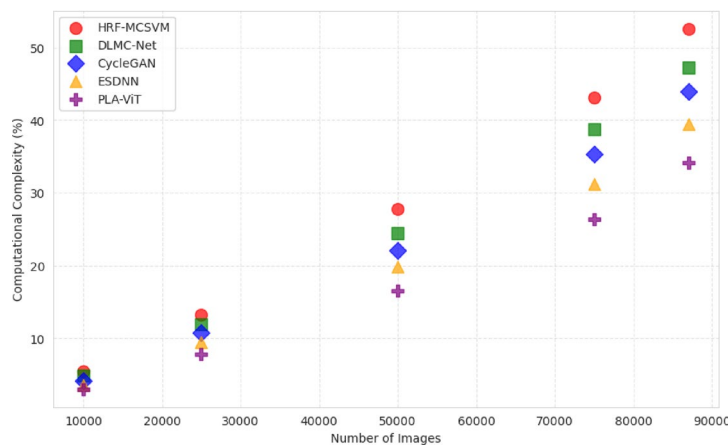


Fig. 9. Computational complexity.

shared characteristics across different tasks. The model relies on attention processes to learn how to prioritize various areas according to their relevance to the current activity. Figure 7 shows the disease localization performance analysis.

Inference time

The model's inference time is a key performance parameter for real-time plant disease detection. Inference time begins when the model receives a leaf snapshot. The model classifies diseases in PLA-ViT at this time. The system must be efficient in inference to be useful in real-world agricultural contexts. This helps agronomists and farmers make decisions faster. An analysis of HRF-MCSVM, DLMC-Net, CycleGAN, ESDNN, and PLA-ViT's inference times shows considerable differences in computational complexity and network design. The simpler structure of traditional machine learning models like HRF-MCSVM reduces inference times. Accuracy is sacrificed. Deep learning-based algorithms like ESDNN and CycleGAN enhance disease localization; however, their advanced feature extraction and deep network layers increase inference time. Any model with fewer weight parameters will always train and infer faster. Thus, PLA-ViT takes less time to infer and train than the existing model. Figure 8 shows the inference time.

Computation complexity

Plant disease detection models' efficiency, scalability, and real-time applicability depend on their computational complexity. A model needs memory, computing power, and execution time to evaluate photos and classify plant diseases. Traditional models like HRF-MCSVM are computationally simple. This is because feature extraction and categorization are simple. However, high-dimensional picture data may make it challenging for these systems to predict complicated leaf disease patterns. Deep learning approaches include adversarial training, convolutional layers, and deeper architectures that may boost classification accuracy. ESDNN, DLMC-Net, and CycleGAN are examples. These breakthroughs demand more complicated computing, larger datasets, and longer training times, which may be expensive. Figure 9 shows computational complexity.

Methods	HRF-MCSVM	DLMC-Net	CycleGAN	ESDNN	PLA-ViT
Accuracy ratio (%)	75.3	82.5	85.6	88.2	98.7
Disease localization performance (%)	72.1	80.3	83.7	86.5	97.4
Inference time (ms)	28	42	35	30	12
Computational complexity	67.2	70.4	65.3	89.7	96.4

Table 4. Comparison analysis.

A comparison is provided in Table 4. The PLA-ViT approach outperforms current plant disease diagnosis methods in accuracy, inference time, computing complexity, and disease localization accuracy. Traditional models, such as HRF-MCSVM, employ fewer computer resources but are less accurate at sickness localization. Deep learning-based techniques like DLMC-Net, CycleGAN, and ESDNN enhance classification accuracy but increase inference time and computing complexity. Low processing load and fast inference time provide PLA-ViT with the best accuracy-efficiency ratio compared to CycleGAN and ESDNN. It is vital for precision agriculture since it can diagnose ailments more precisely than existing methods. Due to its scalability and minimal processing overhead, PLA-ViT is suitable for real-time, large-scale plant disease diagnosis and monitoring.

Conclusion and future enhancements

This study describes the Precision Leaf Analysis with Vision Transformers (PLA-ViT) system for autonomous leaf disease prediction and precision agriculture monitoring. ViT quickly solves agricultural and plant disease image classification problems because of its transformer architecture, originally developed for natural language processing. The model's localization power makes early illness detection easier. A model must learn from common qualities to improve data classification. ViT as a baseline for disease identification in some agricultural species may reduce output loss and enhance crop health. The model can identify various pests so farmers may get early warnings and treat pests promptly, reducing crop losses and pesticide consumption. Farmers may take early action to ensure greater harvests by monitoring plant reactions to abiotic challenges, including drought, salt, and nutrient deficits. The program can correctly determine when crops are at different phases of development, which helps with precision agriculture and conserving resources by optimizing irrigation, fertilizer application, and harvesting.

However, the attention blocks used in this study exhibited poorer performance than the established convolutional blocks across all platforms. Integrating attention block with convolution blocks enhances the models' predictive speed compared to ViT-based models while achieving superior accuracy relative to CNN-based models. The sequence of this amalgamation does not influence forecast speed much, yet utilizing attention block before convolution blocks resulted in improved accuracy. Future research will need object localization networks to identify plant leaves inside extensive images, alongside the resolution of multi-label classification to accommodate various diseases. Alongside identifying a network architecture for multi-label classification and object localization, the compilation of suitably labelled datasets for this issue is acknowledged as a future study in this research.

Data availability

The data used in this research are available in the following links: <https://www.kaggle.com/datasets/vipooooo/new-plant-diseases-dataset>

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Author contributions

Conceptualization, Methodology: MS; Formal analysis and investigation: MS & RG; Writing—original draft preparation: MS; Writing—review and editing: MS & RG; Supervision: RG.

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Declarations

Competing interest

The authors declare no competing interests.

Additional information

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